Where Do I 'Add the Egg'?: Exploring Agency and Ownership in AI Creative Co-Writing Systems

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Abstract

AI co-writing systems challenge long held ideals about agency and ownership in the creative process, thereby hindering widespread adoption. To address this, we investigate conceptions of agency and ownership in AI creative co-writing. Drawing on insights from a review of commercial systems, we developed three co-writing systems with similar functionality but differing interface metaphors: agentic, tool-like, and magical. Through interviews with creative writers (n=18), we explored the role of these metaphors in participants' sense of control and authorship. Our analysis resulted in a taxonomy of agency and ownership subtypes and underscored how metaphorical framings afforded participants different conceptions of agency and ownership. We conclude with recommendations for the design of AI co-writing systems, emphasizing the role of metaphors in participants' creative practice.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **HCI theory, concepts and models**; **Empirical studies in HCI**.

Keywords

AI Writing, Design, Interface Metaphors, LLMs, Writing Assistance

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1 Introduction

The advent of Large Language Models (LLMs) has opened new artistic possibilities for creative writers, resulting in novel forms of poetry, generative novels, and computational literature [44, 77]. This relatively new yet increasingly pervasive form of Artificial Intelligence (AI) has lowered the barrier to entry for amateur or unpracticed creative writers, enabling them to explore genres and forms that may have previously felt inaccessible. Despite the new artistic possibilities brought about by AI, writers' perceptions of these platforms remain mixed, with many skeptical of whether AI can produce sufficiently creative results [10] or whether the introduction of AI into their writing practice would disrupt or threaten their creative integrity [2, 76].

Two of the ways in which writers perceive their interactions with AI co-writing systems are through their perceptions of *agency* when they use a system and their perceptions of *ownership* over the final writing product [26]. A writer's perception of agency is the extent to which they feel a sense of control while using a system, and a writer's perception of ownership is the extent to which they self-identify as an author of the writing co-produced with the AI. To support the broader adoption of AI co-writing systems among writers, it is crucial to consider perceived agency and perceived ownership [40], as these factors can influence a writer's long-term satisfaction with such systems [26].

Previous work has characterized the types of ownership that matter to writers [59] and demonstrated that agency and ownership are generally correlated: writers who feel greater control while AI co-writing generally feel greater ownership over the results [20]. However, little is known about how creative writers conceptualize agency and ownership during interactions with AI creative co-writing systems. This is especially surprising due to creative writing's rich history of deriving multiple conceptualizations of agency and ownership in the creative process: from listening to the muse [65] to post-modern challenges of authorship [3] and aleatory or probabilistic methods of creation [73].

To explore agency and ownership during AI creative co-writing, we used interface metaphors as a design probe. An interface metaphor is a design concept in which familiar objects or concepts are used to represent and facilitate interaction with digital systems, making them easier to understand [45]. Agre's Critical Technical Practice [1] argued that by analyzing the particular metaphors and language of a field of technology, we can identify what the field marginalizes. Interface metaphors are one of the most effective means of altering user perceptions and mental models [9], having previously been used in mobile interfaces and voice-based user interfaces [16, 42]. Our research had two goals. First, we sought to elicit a broader range of conceptualizations of agency and ownership than those currently available in the literature by presented participants with different interface metaphors. Second, we sought to understand how interface metaphors marginalized or afforded particular conceptions of agency and ownership.

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RQ1: What are the different ways writers conceptualize **agency** and **ownership** during AI creative co-writing?

RQ2: What role do **interface metaphors** have in how writers conceptualize **agency** and **ownership** in AI creative co-writing?

To determine which interface metaphors to study, we reviewed popular interface metaphors used in commercial AI creative co-writing systems today. Based on this review and prior literature [40], we identified three archetypes of interface metaphors that were the most prevalent: tool-like, agentic, and magic. We then created three AI co-writing systems, each with an interface feature that exemplified each metaphor.

We then conducted an exploratory study with 18 participants who had creative writing experience. Participants were presented with one of three interface metaphors. Using semi-structured interviews, we explored how participants conceptualized agency and ownership during their co-writing experiences. Our analysis revealed three subtypes of control in AI creative co-writing (i.e., explicit, implicit, and process-based) and three subtypes of ownership (i.e., stylistic, conceptual, and effort-based). Notably, agentic metaphors fostered conceptions of ownership that emphasized the AI's conceptual contributions.

This paper makes three contributions. First, it presents a preliminary review of interface metaphors in common commercial AI co-writing systems. Second, it introduces a taxonomy of agency and ownership in AI creative co-writing. Finally, it offers design recommendations for future AI co-writing systems, highlighting the role of interface metaphors in shaping writers' expectations of control and authorship.

2 Related Work

In this section, we detail research on AI co-writing systems that describes the common dynamics and expectations writers have for these systems and examine conceptions of agency and ownership for users of these systems. We also review research that describes how interface metaphors shape user expectations and interactions with an AI system.

2.1 AI Co-Writing Systems

The idea that computational processes could contribute to writing has a long history, with a very early example being mechanical machines that could generate lines of Latin verse [64]. Early work in Computer Science by Roemmele presented an AI co-writing system for writers that generated suggestions for the next sentences in a story using information retrieval techniques [61]. Later work by Roemmele and Gordon extended this system to use neural networks [60]. A 2024 literature review of intelligent and interactive writing assistants by Lee et al. found that since the mid-2010s, the number of papers published on writing assistants has increased dramatically [40]. Looking specifically at creative writing, there has been research on new techniques to support story writing [33, 80], playwriting [46], character development [56], prewriting [75], nonlinear story writing (also known as 'multiverse') [27], worldbuilding [14], cartoon-captioning [35], and feedback generation [4]. Such research has demonstrated that AI has the potential to support a variety of creative writing activities and aspects of the creative writing process. However, such research remains in its early stages and is largely demonstrative, as these systems have yet to see widespread adoption among writers and are typically studied in small lab-based settings.

In addition to systems that support specific creative writing activities, there has also been research investigating how creative writers interact with, and reason about, writing support. CoAuthor provided a dataset to study AI co-writing and demonstrated how tracing writing interactions can allow for rigorous investigations into human-computer collaboration [41]. Other research has investigated how creative writers incorporate AI suggestions, for instance, engaging in 'integrative leaps' [68] or viewing suggestions antagonistically [7]. Gero et al. reported on how writers reasoned about AI support differently than human support [26] and Kim et al. identified the benefits of AI co-writing based on the dynamics between an author, artifact, and the audience [37]. Such research has also outlined several issues that arise during AI co-writing, including a decreased sense of agency and ownership, which we consider next.

2.2 Agency and Ownership with AI Co-Writing

Early work on understanding AI co-writing in creative contexts suggested that AI involvement may decrease writers' sense of ownership [24, 25]. More recent work on AI co-writing for creative and argumentative texts explicitly investigated issues of agency and ownership, finding that the less a user writes, the less ownership they felt over the final text [41]. Others have found that AI assistance for idea generation decreases feelings of ownership [29] and that suggesting paragraphs of text may increase writing quality but at the cost of decreased feelings of ownership [18].

Other research has investigated additional factors that impact perceptions of ownership. For example, Kadoma et al. found that the style of AI suggested text (e.g., hesitant or assertive) can impact writers' perceptions of ownership [34]. In a two-week diary study, Kobiella et al. investigated young professionals' perceptions of ChatGPT and found that feelings of ownership were related to how much they felt their contribution was central to the task at hand [39]. While investigating the relationship between authorship and ownership, Draxler et al. found that higher levels of participant influence on text increased one's sense of ownership, ownership and authorship were not always correlated, and that participants were more likely to claim authorship of AI-assisted texts than human-assisted ones [20]. Reza et al. recently identified a distinction between academic and creative writers' values in ownership, finding that academic writers valued the ideas and content of pieces they created, while creative writers valued the formal and linguistic elements [59]. Hoque et al. designed a system to encourage agency and ownership that tracked the provenance of text and found that provenance information increased a writer's sense of agency and ownership by supporting active considerations of what the AI contributed [31]. Such work demonstrates that perceptions of ownership were not only a function of how much text was generated by an AI assistant but were also influenced by perceptions of the AI as a collaborative partner and judgments about its contribution.

2.3 Interface Metaphors for AI

Users may have preconceived notions about AI systems, so metaphors can help users understand the capabilities of an AI agent by proposing an alternative mental model [12, 26]. The mental models evoked by a metaphorical comparison facilitate users' familiarization with a system by creating a set of expectations around its functionality [17]. By incorporating different metaphors into AI systems, designers can create shared meanings that help users express nuanced views about, and experiences with, an AI agent [17].

Comparative research on metaphors for AI has demonstrated the impact metaphors can have on user interactions with task-based AI systems, e.g., AI that assists with well-defined tasks such as travel planning [36] or waste sorting [52]. For example, Khadpe et al. explored agentic metaphors that portrayed chatbots as different human personas and found that they changed user satisfaction with a system even when the system's capabilities were unchanged [36]. Users have also been found to project their own metaphors onto AI, which influenced their interactions [12, 52]. Chin and Desai investigated older adults' use of voice assistants and found that they ascribed a variety of object- and human-based metaphors, thus suggesting they had different conceptions of the assistant's agency, autonomy, and animacy [12]. This comparative research has shown that interface metaphors, whether explicitly assigned by designers or implicitly adopted by users, significantly influence users' perceptions and interactions with AI systems.

In the domain of AI co-writing systems, interface metaphors have also been found to impact the types of interactions that users expect to encounter. Based on a review of prior literature, Lee et al. described three *interaction metaphors* for AI co-writing systems [40]. "Agent" metaphors described designs that used anthropomorphized AI, with conversational interactions and streaming text that simulated an interaction with another person. "Tool" metaphors compartmentalized the capabilities of an AI system and presented controls using familiar interface elements such as buttons and dropdown menus. Systems designed with "hybrid" metaphors mixed elements from "agents" and "tools." Interaction metaphors, as used in Lee et al., were closely related to interface metaphors and were often deployed in concert to present intuitive affordances to users. While interface metaphors relate visual and organizational design to other objects through metaphors, interaction metaphors relate the actions a user can perform. Lee et al.'s use includes both interactions (e.g., conversational interactions) and interface presentation (e.g., streaming text), so their findings are relevant to our study of interface metaphors.

In addition, Biermann et al. used a design workbook approach to compare different AI interfaces and investigate how writers' personal values affected their intent to adopt AI in their practices [6]. Their research revealed differences in the preferred divisions of control between writers and AIs based on the interface presented and the writer's writing process, professional status, feelings of self-efficacy, and emotional attachment to their prose. Our research extends this prior work by using the demonstrated perceptual impacts of AI interface metaphors to modulate and examine how writers conceptualize agency and ownership in their interactions with AI co-writing systems.

3 Interface Metaphors in Commercial AI Creative Co-Writing Systems

To facilitate the design of interface metaphor systems that would closely mimic the conditions that writers are most likely to encounter, in May 2024, we conducted a review of the interface metaphors that were most often used within commercial AI co-writing systems. Our review methogology was based on prior HCI research methods that compared commercial systems [5].

As multiple metaphorical forces can have a combined effect on the interaction process, identifying and investigating each metaphor individually enabled us to develop a more detailed understanding of how each metaphor could influence the users of commercial interfaces. Additionally, by identifying archetypal metaphors from which hybrid metaphors are most commonly composed, we were able to map the design space of interface metaphors in AI creative co-writing systems and explore how different interaction dynamics in AI creative co-writing could elicit different conceptions of agency and ownership.

Our review began with a preliminary Google search for popular AI co-writing systems that resulted in 45 AI co-writing systems. We then categorized the systems according to their intended audiences and use cases, using their marketing materials as reference. We excluded systems that were not targeted towards creative writers, leaving a total of nine systems. We then conducted additional searches within the r/writingwithai[1] and r/writing[2] Reddit subreddits which revealed the popularity of ChatGPT, Google Gemini, and Grammarly. These three systems were then included in our review, resulting in a total of 12 AI creative co-writing systems being reviewed: ChatGPT, DreamGen, Fictionary, Google Gemini in Google Docs, Grammarly, NovelAI, NovelCraft, Novelcrafter, Quillbot, Squibler, Sudowrite, and Writesonic.

Our review included an hour-long exploration of all the features and interface elements within each system. We captured screen recordings of each system and recorded the textual metaphors presented in text descriptions and iconographic metaphors in icons representing AI assistance. We then characterized each system based on the interface presentation they used (e.g., icons and descriptions), the interaction models they afforded to the user, and the user agency they enabled or constrained.

Using Lee et al.'s framework [40] as a guide, our analysis and use of the 12 systems revealed that many systems had clusters of interface elements, presentation methods, and methods of interaction that corresponded to Lee et al.'s descriptions of "Tool" and "Agent" interfaces. We also identified a cluster that was not captured by Lee et al.'s framework, i.e., a *magic-like* archetypes which had distinctive magical interface presentations and simplified interaction designs. While some systems [57, 69] often instantiated several interface metaphors for each of its AI interfaces, most interface elements reflected a single interface metaphor, so we chose not to include the hybrid metaphor as an archetype.

Thus, our review revealed three recurring archetypes of how AI interactions were presented to users through interface elements: Agent-like, Tool-like, and Magic-like (Table 1). While the archetypes are not an exhaustive categorization of all possible AI interface

[1] https://www.reddit.com/r/WritingWithAI/
[2] https://www.reddit.com/r/writing/

Table 1: The three archetypal interface metaphors found in commercial AI co-writing systems today.

Archetype	Characteristics	Interface Elements	AI Interaction	User Agency	Examples
Agent-like	Personified AI, collaborative partnership, natural language interface	Chat windows, streaming text, persona icons, first-person pronouns	Conversational, iterative, often separated from main text	Shared agency, AI as collaborative partner	ChatGPT, DreamGen, Novelcrafter
Tool-like	Precise control over AI capabilities, mechanical imagery, compartmentalized functions	Buttons, drop-down menus, sliders, keyword inputs for length, tone, and style	Direct manipulation, integrated with text editing	Fine-grained user control, AI is an instrument extending user's creative agency	Squibler, Writesonic
Magic-like	Simple automatic functions, magical iconography, emphasis on seamlessness	Wands, sparkles, minimal controls, one-click generation	One-shot automation, minimal iteration	Minimal explicit control, automatic "enchanted" generations	NovelCraft, Sudowrite (Auto Write)

metaphors, they emerged as the most prevalent approaches found in current commercial creative co-writing systems. Summary statistics of the review can be found in Appendix A. Following the identification of the three archetypes, we conducted a supplementary survey of HCI research to assess whether these metaphors also appeared in research prototypes. Our survey confirmed these interface metaphors were present in the HCI literature as well, but further work is needed to perform a systematic review of these research prototypes as they were not publicly available at the time of our review.

3.1 Interface Metaphor Archetypes

The **Agent-like archetype**, found in seven systems was primarily implemented through conversational interfaces with a personified AI that could accomplish a variety of functions. Some systems enhanced this sense of agency through specific personas applied to the AI, such as Novelcrafter's support AI Nia, which was depicted as a cute goblin [50]. These interfaces emphasized natural language interaction and often presented the AI as a collaborative partner that addressed the user and contributed directly in the writing area. This archetype was typified by ChatGPT (Figure 1a), which featured a chat interface that streamed text that mimicked the AI typing and framed the AI as a personified agent by using first-person pronouns and directly addressing the user. DreamGen [21] employed a similar streaming text pattern to reinforce its agent-like presence. Agent-like interfaces were also common in HCI literature on AI co-writing, in both creative writing [24, 47, 56, 62] and other writing contexts [11, 13, 53, 66, 71, 74], featuring chat interfaces with avatars, personified agents, and streaming text that was similar to the reviewed commercial systems.

The **Tool-like archetype**, present in 10 of the 12 systems, offered precise control over AI capabilities and compartmentalized functions to diminish a user's perception of independent AI agency. These elements used mechanical imagery such as pencils or gears, and provided interface elements for users to constrain the length, tone, and style of AI-generated content (e.g., buttons, dropdown menus, sliders, and keywords). For example, Squibler's Smart Write interface (Figure 2a) provided explicit control elements such as sliders for the amount of dialogue in the text, a text box to define the content of the generated section, and a dropdown menu to select the narrative perspective. Writesonic's Continue Writing feature [78], which included interface parameter controls, also exemplified this approach. HCI research has explored many designs of tool-like interfaces in AI co-writing, with workflows and controls that extended beyond those we observed in commercial systems. Many interfaces resembled commercial instantiations of this archetype, utilizing buttons, dropdown menus, and keywords (in creative writing [51, 58, 70, 80] and other writing contexts [28, 38, 63]). Others used tool-like interfaces that were not found in the commercial systems, such as a tool for the generation of stories containing target words for language learning [54], and hierarchical generation [46].

The **Magic-like archetype**, present in nine systems, did not fall along the Tool-Hybrid-Agent spectrum described by Lee et al. [40]. Unlike the other interface metaphors, there was not a single system that instantiated a complete Magic-like interface, however, we observed a consistent pattern where systems presented automatic, simplified functions and sub-interfaces using magical framing and iconography including wands and sparkles. While decidedly not agentic, these components failed to offer the fine-grained control of tool-like interfaces. Instead, they were distinguished by their simplicity and lack of precise control over the AI output. Sudowrite's Auto Write feature (Figure 3a) emphasized automaticity and limited user controls compared to other systems, but had textual and iconographic references to magic. NovelCraft [49] reinforced magical framing through wand iconography, while sparkles (which have been found to contribute to the perception of magic [32]) appeared in 8 of the 12 systems. The application of magic metaphors to AI has been previously explored in HCI and Design research [43], highlighting the mysteriousness of the actual functioning of a system, enchantment, and seamlessness as core design principles. We observed the application of magic metaphors in multiple VR and visual art co-creative AI systems [23, 30, 79, 81]. While the designers of these systems emphasized the magic systems' immersiveness and seamlessness [23, 30, 81], and used magical names and iconographic framing, we did not find any research using magic-like interfaces or framing in AI co-writing systems. This absence is particularly striking given the widespread use of this magical framing in commercial writing tools, suggesting that magic metaphors represent a promising yet underexplored direction for AI co-writing research.

Table 2: Demographics of the Exploratory Study Participants.

Participant	Gender	Occupation	Group	AI Interface
1	M	Novelist, Playwright	Professional Creative Writer	Autowrite (Tool-like)
2	M	Novelist, Short Story Author	Professional Creative Writer	Autowrite (Tool-like)
3	M	Poet, Short Story Author	Professional Creative Writer	Wordsworth (Agent-like)
4	M	Novelist, English Professor	Professional Creative Writer	Magic Quill (Magic-like)
5	M	Novelist, Short Story Author	Professional Creative Writer	Magic Quill (Magic-like)
6	F	AI Consultant	Non-Professional Writer	Autowrite (Tool-like)
7	F	PR Consultant	Non-Professional Writer	Magic Quill (Magic-like)
8	F	Short Story Author	Professional Creative Writer	Magic Quill (Magic-like)
9	M	Comedy Writer	Professional Creative Writer	Wordsworth (Agent-like)
10	F	HCI Researcher	Non-Professional Writer	Magic Quill (Magic-like)
11	F	Animation Researcher	Non-Professional Writer	Wordsworth (Agent-like)
12	F	Film Producer	Non-Professional Writer	Magic Quill (Magic-like)
13	M	Engineering Student	Non-Professional Writer	Wordsworth (Agent-like)
14	M	Poet, Short Story Author	Professional Creative Writer	Autowrite (Tool-like)
15	M	Communications Student	Non-Professional Writer	Wordsworth (Agent-like)
16	M	Educator	Non-Professional Writer	Autowrite (Tool-like)
17	M	Prop Master, Art Director	Non-Professional Writer	Autowrite (Tool-like)
18	F	Novelist, Short Story Author	Professional Creative Writer	Wordsworth (Agent-like)

4 Exploratory Study of Interface Metaphors for AI Co-Design Interfaces

We conducted an exploratory study with professional and non-professional creative writers to understand their conceptions of aspects of agency (e.g., ease and precision of control over the system, control over the writing process, and whether the system supported and aligned with the writer's goals) and ownership (e.g., authorship, collaborative authorship, the representation of the writer's unique style and voice, etc.) when using AI co-writing systems. Participants first completed a short creative writing task using a prototype framed with one of three interface metaphors. They then participated in a semi-structured interview with two researchers. During the interviews, participants were asked how their collaboration with the AI could change if the generated text was less similar to their voice or style, if their interaction with the AI surprised them, and if they felt that the AI was a co-author of the resulting output. The study was approved by our institution's Research Ethics Board and each participant provided their written and verbal consent before starting the study.

4.1 Participants

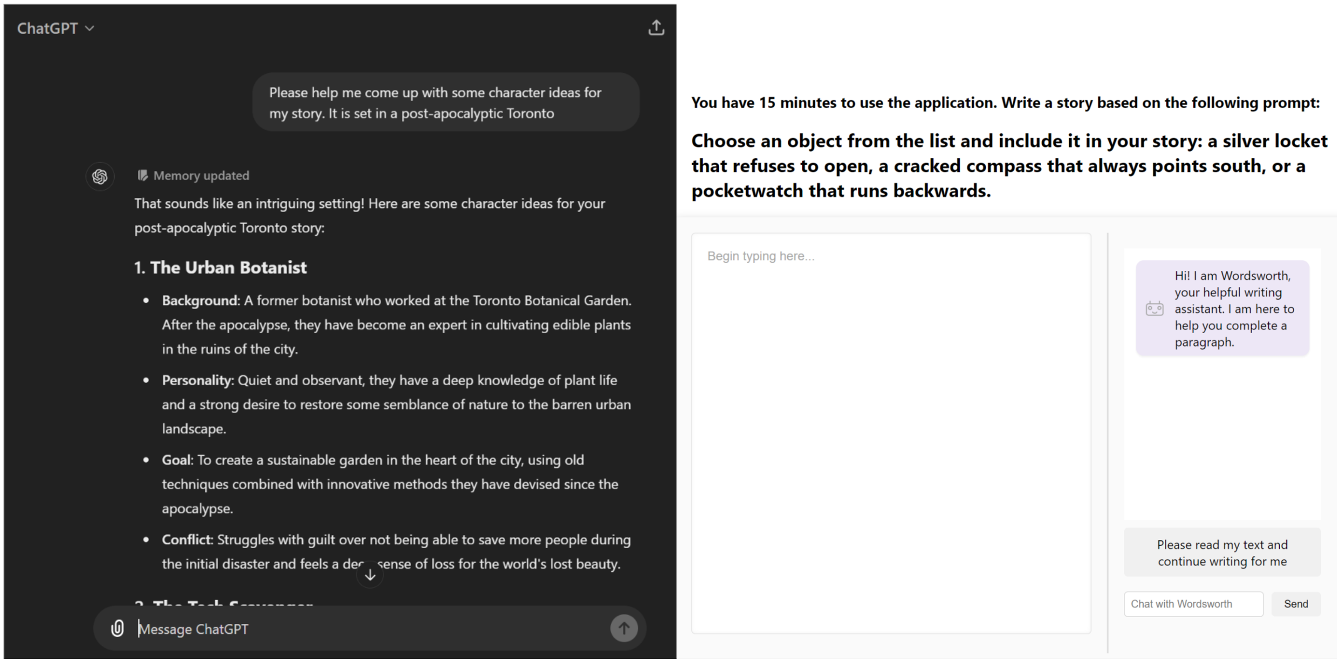
We recruited 18 participants using the first author's network of fellow literary writers, social media, and outreach within our institution. We targeted two distinct groups: professional creative writers (n=9) and non-professional creative writers (n=9). While the distinction between the groups was relevant while analyzing the data for (RQ1), it was not relevant while analyzing the data for (RQ2). Therefore, when analyzing the data for (RQ2), participants were combined into a single group (n=18). Eleven participants self-identified as men and 7 participants self-identified as women.

Our study recruited professional creative writers from the North American English-language literary fiction community, a community that aims to create works of artistic merit and cultural importance. In English fiction, such writers generally emphasize precision of language, style, and nuance of character. We chose to study these writers because their interest in precise style and deep consideration for various modes of creative agency and authorship makes them distinct from the general population. The non-professional writer group was included to provide a contrasting perspective and to explore how non-professional writers might perceive agency and ownership during AI co-writing. The study took between 60 and 90 minutes per participant. Participants received a $45 CAD Amazon gift card as an honorarium for their participation.

4.2 AI Creative Co-Writing Systems

To explore how participants reasoned about agency and ownership in AI creative co-writing systems, we developed three prototype AI co-writing systems: Wordsworth (agent-like), Autowrite (tool-like), and Magic Quill (magic-like). The prototypes were designed with functionality to support our investigation of the role of interface metaphors in how writers conceptualize agency and ownership (RQ2). By keeping the functionality consistent across all three prototypes (i.e., limited to text completion), our quantitative results thus stemmed from the use of each interface metaphor in isolation. Each prototype was presented to participants with a name that reinforced its metaphorical framing.

Our web-based prototype systems used React for the front-end, Flask for the back-end, and was hosted on Heroku. All AI-generated text within each prototype was generated by GPT-3.5-turbo via OpenAI's API. We utilized the default parameter values (as of June 2024) for all parameters besides model, prompt, and max tokens. The model was chosen as it offered sufficient consistency in text understanding, completion capability, and processing speed to support our focus on effects of the interface metaphors. Completions were generated by GPT-3.5-turbo via OpenAI's API and were limited to 250 tokens (i.e., approximately 100 words). They were appended to the end of the editing area on the left side of each interface as a completion of the existing content. Each completion was preceded by the system prompt: *"Suggest a continuation for this prompt. Your reply should only be the continuation. The continuation should*



(a) Agent-like Commercial Reference Example: ChatGPT

(b) Wordsworth Interface

Figure 1: (a) A commercial example of an agent-like interface metaphor: ChatGPT's conversational interface used a chat window to communicate with the user. (b) The Wordsworth interface had a chat window in the right panel for conversational interactions with the AI, presented using a stylized robot icon.

be grammatically correct and flow naturally from the prompt. The continuation should be paragraph-length."

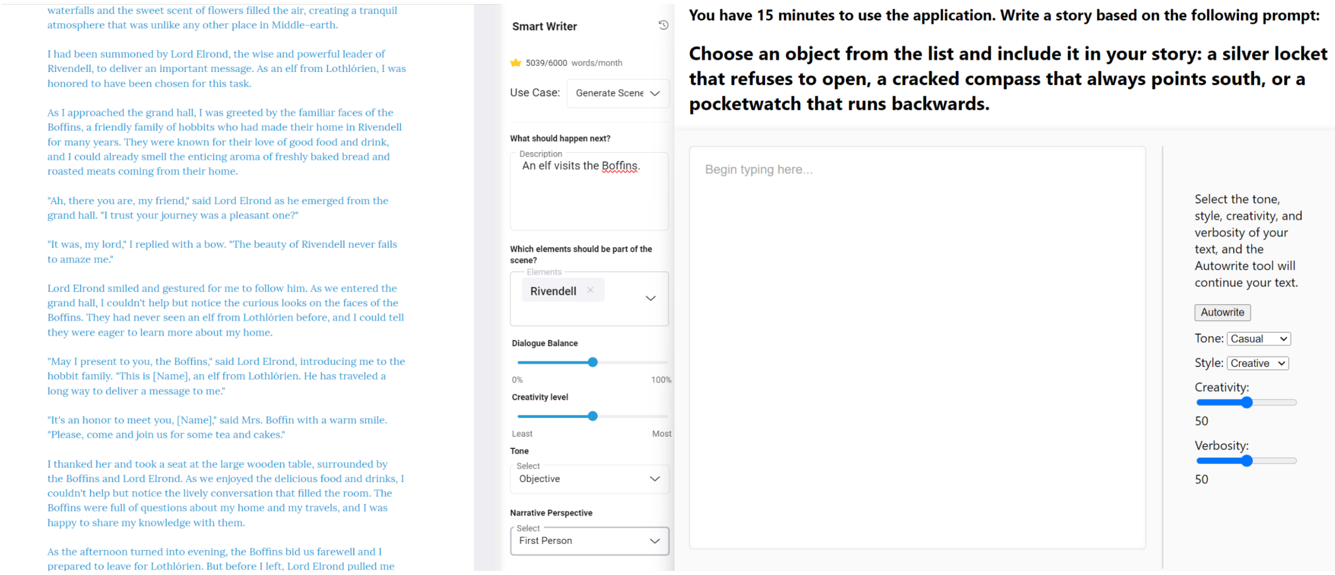
We varied the *presentation* of the same text completion functionality within each prototype through the interface elements and iconographic framing identified in Section 3's review of commercial AI co-writing systems. Each prototype system instantiated different patterns observed in the three archetypes we identified (Table 1), which are described in detail below. The differences within each prototype allowed us to explore how different interface metaphors, distinct from different AI capabilities, shaped writers' experiences of agency and ownership.

Wordsworth. Wordsworth instantiated the agent-like archetype, which our commercial review found to be primarily characterized by conversational interfaces, personified AI presentation, and streaming text that simulated typing. To instantiate these patterns, Wordsworth featured a chat interface with a stylized robot icon and an icon of a human waving to reinforce Human-Agent collaboration (Figure 1b). Similar to commercial agent-like systems such as ChatGPT, the system streamed text character-by-character into the text area, reinforcing the sense of an agentic partner composing alongside the participant.

In service of our goal to vary presentation while keeping functionality consistent between prototypes, Wordsworth's AI output in the text area was limited to basic text completion and the system could not accommodate requests for auxiliary functions like research. GPT-3.5-turbo parsed the user's natural language queries in the chat interface to identify requests for completion, but all user-facing responses employed scripted responses rather than AI-generated text. Based on our initial testing, we found that natural language requests for completion introduced significantly more interaction friction by requiring users to type a full message compared with our other button-based prototype designs, so a one-click button labeled *"Please read my text and continue writing for me"* automatically entered and sent this request, presented as a message, to Wordsworth. This button ensured that participants could interact with a similar level efficiency as the other prototypes, while still presenting the request within the framing of a chat in natural language, and emphasizing the personified capabilities (i.e., reading and writing) of the AI.

Autowrite. Autowrite instantiated the tool-like archetype, which our commercial review found emphasized precise control through explicit interface elements such as dropdown menus, sliders, and parameter fields. Commercial tool-like systems featured mechanical imagery and compartmentalized functions that diminished perceptions of independent AI agency, instead positioning the AI as an instrument granting fine-grained user control. To instantiate these patterns, Autowrite provided dropdown menus for "Tone" (e.g., Casual, Formal, Academic) and "Style" (e.g., Creative, Technical, Narrative), along with two sliders for "Creativity" and "Verbosity" that were adjustable from 0 to 100 (Figure 2b).

This system was similar to Squibler's Smart Write and Writesonic's Continue Writing feature, which offered parameter controls to give participants a sense of precise manipulation over AI output, however, the provided controls did not actually modify the AI's



(a) Tool-like Commercial Reference Example: Squibler

(b) Autowrite Interface

Figure 2: (a) A commercial example of a tool-like interface metaphor: Squibler's Smart Writer interface used control elements such as dropdown menus and sliders to support fine-grained control over the generated text. (b) The Autowrite interface had dropdown menus and sliders in the right panel to provide the illusion of fine-grained control over the generated text.

output. This deception was necessary to maintain similar functionality across all three prototypes while preserving the tool-like metaphor's characteristic emphasis on user control and explicit parameterization. Unlike Wordsworth's streaming text, which emphasized composition, Autowrite delivered the complete AI-generated text approximately one second after clicking the "Autowrite" button. This immediate delivery pattern reinforced the tool metaphor: the AI functioned as an instrument that executed user-specified operations rather than as an agent that gradually composed text. The combination of exposed controls, mechanical button interaction, and instant delivery created a distinct interaction dynamic that positioned the participant as operator and the AI as a controlled, predictable tool.

Magic Quill. Magic Quill instantiated the magic-like archetype, which our commercial review found to emphasize automaticity, minimal user control over AI output, simplicity, and obscurity of the underlying AI process. These design choices positioned AI assistance as mysterious, effortless, and seamless. To instantiate this archetype, Magic Quill featured a single button displaying a quill icon with a continuous subtle sparkle animation (Figure 3). The sparkle animation referenced sparkle motifs from Sudowrite, DreamGen, Grammarly, and other commercial systems, and previous work on perceptions of magic in interaction design. [32]

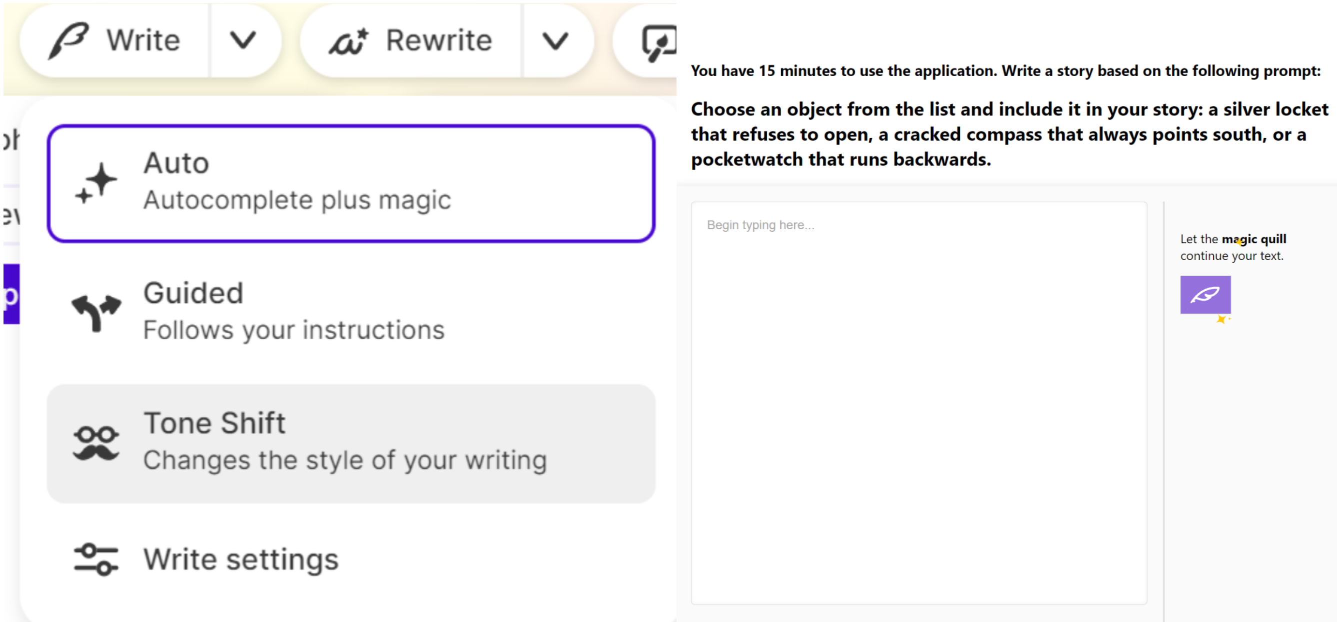
When the quill icon was clicked, AI-generated text faded into the text area and sparkle animations that intensified over a period of one second appeared across the interface. This fade-in effect was designed to evoke 'enchanted' writing from popular culture, such as Tom Riddle's diary from Harry Potter.[3] The single-click invocation pattern mirrored those found in commercial systems like Sudowrite's Auto Write feature, which emphasized automaticity and limited controls compared to tool-like alternatives. Deliberate simplification and the presentation of enchantment led Magic Quill to position the AI as a source of enchanted assistance rather than a controllable instrument or collaborative agent.

4.3 Procedure

The exploratory study was conducted online via Zoom and transcribed by Otter. At the beginning of each session, the researchers obtained consent from the participant to record and transcribe audio. Any concerns about data management or study procedures were addressed before proceeding.

Introductory questions were then used to establish rapport with the participant and gather information about their writing processes and experiences with AI tools. These questions explored typical writing routines and familiarity with, or attitudes towards, AI co-writing systems. After the introductory questions, participants were remotely introduced to one of the three AI co-writing systems. Our study used a between-subjects design, but our goal was exploratory and descriptive rather than confirmatory. We aimed to characterize how participants experienced each metaphor archetype, providing empirical grounding for future controlled comparisons with quantitatively demonstrative effects. The conditions were randomly assigned while ensuring an even distribution across the two groups of writers. The research team explained the prototype and allowed

[3]An examples of fading in enchanted text from the Harry Potter and the Chamber of Secrets movie: https://www.youtube.com/watch?v=eh4b5zC0sB4



(a) Magic-like Commercial Reference Example: Sudowrite **(b) Magic Quill Interface**

Figure 3: (a) A commercial example of a magic-like interface: Although none of the reviewed systems instantiated a complete Magic-like interface, Sudowrite's Write button menu used a sparkle icon, the text "Autocomplete plus magic", and withheld other interface controls to invoke magic as a metaphor for creative generation. (b) The Magic Quill interface featured a single point of control, animated sparkles around the quill button, and the text "magic quill."

participants to familiarize themselves with its functionality. Participants then engaged with the system for 15 minutes while they were observed by the research team. While a writing prompt was provided, participants were free to choose their own topic and use the AI as much or as little as they preferred.

Following their usage of the prototype, participants were asked about their overall impressions, perceived changes in their writing process, and the potential integration of such tools into their regular practices. These questions captured immediate reactions and broad perceptions of each condition. Participants were then asked how the interface supported or interfered with their sense of control, if it influenced the direction of their writing, if it affected their perception of authorship, and whether the final product represented their unique style and voice (RQ1). Participants' conceptions may have been influenced by prior AI experience or literacy, media exposure, age, or other factors that can inform a user's mental model of AI systems. However, our study aimed to understand what conceptions of agency and ownership were afforded by interface metaphors, not to exhaustively understand all sources of influence. Follow-up questions then used speculative comparative scenarios to understand the role of the presented interface metaphor in the perception of agency and control (RQ2). Proposed changes included generating different lengths of output, providing more or fewer explicit controls like dropdown menus or sliders, or writing more seamlessly in the writer's style or voice. Participants were asked if these changes affected whether or not they felt like they collaborated with the AI and how in control they felt during their interaction with the AI. The complete study protocol can be found in (Section 9).

4.4 Measures and Analysis

The analysis began with an initial open coding of the study transcripts provided by Otter to identify recurring concepts in participants' responses. This process involved listening to each interview and assigning codes to relevant text segments to capture the essence of participants' experiences and perspectives. Following this initial coding, we conducted a reflexive thematic analysis, as described by Braun and Clarke [15], to develop and refine themes from the coded data. Coding was performed over a two week period and was performed by two members of the research team. Both coders coded the data individually, developed preliminary themes, and then reviewed the codes together to further develop the themes and ensure that they were empirically justified.

The research team identified two sets of themes in the data. The first was the range of conceptualizations of agency and ownership that were exposed by the design probe with the prototype systems. After identifying recurring conceptions of agency and ownership, we constructed a taxonomy based on these recurring conceptions. This taxonomy addressed (RQ1) but did not speak to the role of interface metaphors in conceptualizing agency and ownership. The second set of themes suggested what role particular interface metaphors had in particular conceptions of agency and ownership. These themes answered (RQ2) and spoke to the role of metaphors in conceptualizing agency and ownership.

While the themes from the professional and non-professional writers were analyzed separately for (RQ1), when analyzing for (RQ2) no differences relevant to the analysis were found between the groups. Thus, the research team concluded that participants' levels of professionalism did not affect how interface metaphors entangled with their conceptions of agency and ownership and merged both groups into a single group for a secondary analysis. The original themes were then reviewed and refined through discussions between the two coders to arrive at the sets of themes presented herein.

5 Exploratory Study Results

The findings reflected the nuance and complexity of participants' conceptions of agency and ownership when interacting with AI co-writing systems, and how those conceptions evolved in the presence of different interface metaphors. Below, we first describe the points of control and types of agency and ownership that were described by participants across metaphor conditions (RQ1), and then discuss which conceptions of agency and ownership were afforded by agent-like and tool-like metaphors (RQ2).

5.1 Conceptions of Agency and Ownership (RQ1)

In this section, we describe the three conceptions of agency and ownership that arose during our study.

5.1.1 Conceptions of Agency. In our study, participants expressed a desire for *explicit, implicit,* and *writing process* agency.

Explicit Agency. Some participants felt that their sense of agency was rooted in their explicit control over their interactions with the AI via interface elements outside the main writing area, such as buttons or dropdown menus (P2, P6, P7, P10). For these participants, choosing when and how they asked the AI to generate text afforded them more control: *"I feel better in general if I have more control over my work. So when I change the parameter that means if I like it, I can make it better by controlling that parameter. If I don't like it, then I can remove it or I can adjust the parameters. I think the flexibility leads the user to feel more confident to use the tool."* (P6).

Implicit Agency. Other participants used a form of *implicit agency* during their interaction, i.e., they "prompted" the AI with their previously written text (P1, P6, P17, P15, P18). As P1 noted, *"I kept trying to get it to generate a different tone, right, and to go down a different path and I felt like it took a lot of prompts."* Given that all 18 participants had the option of using the previously written text to guide the AI regardless of the system they used, it is surprising that only 5 mentioned this implicit guidance. Further, the use of the word "prompt" suggests a familiarity with existing agent-like metaphors for thinking about AI, even though three of the five participants that mentioned prompting used the tool-like interface. This suggests that preconceived metaphors of AI from past experiences may have influenced how participants perceived these systems, into addition to AI literacy, cultural backgrounds, media exposure, or any number of other factors beyond the scope of this study. Future work will need to explore how these other factors may inform conceptions of agency and ownership.

Writing Process Agency. For others, the ability to continue using traditional writing processes resulted in a sense of *writing process agency.* These participants were able to find an internal point of control in their writing process rather than one within their interaction with the AI. For instance, participants whose writing process involved simultaneous editing and drafting did not feel that their sense of control was threatened by the AI because their usual writing process was not altered, e.g., *"I was obviously still in control of the piece, because I actually deleted a few sentences ... I kept the parts I liked, I deleted the parts I didn't like, and I could still edit them."* (P11). P7 described editing as an inevitable part of working with the AI generated text and conceptualized their role as that of a senior partner or editor. For these participants, maintaining control during the writing process was easy regardless of the system they used because they could still use their traditional writing process.

Although control via editing was the most frequently referenced form of writing process control, other sources of alignment and misalignment between participants normal processes and AI co-writing processes were also observed. P1 described their process as *"...a sort of invitation to my subconscious to surprise me, to try and access some stuff that my conscious mind isn't able to access well"*, and therefore felt that large chunks of output from the AI violated this ability to surprise themselves. P2 described their process as *"aleatory and improvisational,"* and described struggling to stay in that familiar process during their interaction, i.e., *"Every time it generated something it was a bit like 'oh, I wouldn't write that, I wouldn't go that way' but I still want to keep going. I want to see if it could give me something that I wanted, but obviously without me telegraphing my intentions outright."* Participants with processes relying on randomness or reduced active intent were more willing to cede some control of their process. They seemed to need an AI collaborator that was receptive to their goals and style without forcing them to control it in ways that required clear intent and thus violated their writing process.

5.1.2 Conceptions of Ownership. As with agency, participants reported several ways of conceptualizing their ownership over the resulting writing. P1 said *"It feels sort of like in the 1950's with [store bought] pancake batter. All you had to do was add water but no one wanted to use it because it felt like they were cheating. So what [the pancake company] did is re-advertised that you had to add an egg to it. You didn't actually have to add the egg, but the act of adding the egg was just enough for people to feel like it had their own homemade touch."* In this section we explore how and when participants felt they had added their "egg" to the AI generated text.

Conceptual Ownership. One notable difference between the professional creative writers and the non-professional writers was that each tended to derive ownership from different parts of the process. Generally, non-professional writer participants tended to believe that conceptual contributions or "ideas" were the more important contribution and provided a sense of *conceptual ownership* (P7, P16, P10, P11, P13). P11 stated that *"When you're writing a book, you can have an editor, but still, the ownership and the authorship belongs to a person who's writing a book, because they contribute the ideas and they are the creative brain behind the book."* As long as the AI co-writing system was expanding on the ideas that they provided, these participants felt that they had more ownership over the draft.

Stylistic Ownership. Professional creative writer participants, in contrast, tended to place more emphasis on style, language, and the words themselves. P1 noted that, "*I think there's a difference between feeling ownership over syntax and style versus feeling ownership over content, right?*" P1 felt that the generated text was so disparate from their own style, they could not claim any ownership over the final product. After using our prototype system, P18 no longer considered themselves the author of the piece because the AI was inserting language directly into the text. P18 described the final piece as "collaborative" and compared the process to the exquisite corpse collaborative writing game[4]: *"It didn't feel like a solo thing. It didn't feel like I was the writer at all times as I wasn't. Like that little game that you play when you're kids, where you say one sentence and then you have to repeat the sentence, and then add on to the sentence, and then you keep going with the story down the line."* When asked how to assess whether they or the AI had greater ownership over the text, some of these participants emphasized language over ideas by suggesting that comparing the number of words that the AI composed to the number of words that the writer composed would be a reasonable way to assess ownership. For example, P8 stated, *"Maybe it's something to do with the amount of words taken without modification. If I had grabbed this much text and kept that I might feel like I had less of a hand in it."*

Effort-Based Ownership. Many participants said that they would not take ownership over a draft until they had spent a substantial time editing it, which is a form of *effort-based ownership* (P1, P2, P14, P12). For example, P14 noted, *"It's that it's doing the cognitive work, it would feel like it has sort of done my thinking for me. Maybe this is a thing I've internalized — Calvinism or something. It's like you have to suffer... I have to say that I did this, that I sweated it out."* For this participant, agency and ownership were interrelated because if they exerted energy on points of control they considered important, then they were more likely to claim ownership over the resulting text.

5.2 Interface Metaphors' Role in Conceptualizing Agency and Ownership (RQ2)

Below, we describe the role of two of the three prototype systems in forming conceptions of agency and ownership (RQ2). We first describe how participants presented with the tool-like metaphor found points of control in tool-like aspects of the interface. We then describe how participants presented with the agent-like metaphor tended to associate ownership with conceptual contributions.

Notably no consistent role was found for the magic-like interface. One might speculate that the ambiguous nature of the magic metaphor would have reinforced an implicit sense of control in which the participant was unsure how the AI was magically extending their text. In practice, however, our prototype used a single-button interface consistent with how magic appears in current commercial AI co-writing interfaces. Participants may therefore have partially associated the interface with past experiences with having explicit agency over computer interfaces, leading to a mixed orientation. We discuss this further in Section 8.

[4]https://en.wikipedia.org/wiki/exquisite_corpse

5.2.1 Points of Control Found in Tool-Like Interface Elements. In this section, we describe a pattern discovered among participants who used the tool-like metaphor (Figure 2b): they gravitated toward conceptions of agency formed through tool-like interface elements.

P2, P6 and P16 stated that they needed to have explicit control over the AI and stylistic ownership over its output to feel that the writing process was cooperative. As P16 stated, *"If the AI writes seamlessly [in my style] then I'll feel very good about it, and a collaboration is going to be equal."* Likewise, P2 said that while they would tend to not use an AI co-writing system, *"the more control that I can have with regard to a program like this, it lowers my resistance, shall I say, to using it."*

Participants who used the tool-like system viewed the dropdown menus and sliders that "controlled" the style, verbosity, or creativity of the generated text as important points of control (P1, P2, P6, P14, and P16). P6 stated, *"I do feel better in general because I have more control over my work so when I change the parameter, that means, if I like it, I can make it better. I think the flexibility leads the user to feel more confident to use the tool."* P2 noted *"I think I would go down the rabbit hole and just play with every parameter and that's what it would take for me to kind of accept the program. I like fine grain control, wanting to see control or accept control."* For some participants, the illusion of explicit control made them feel greater agency, while others who noticed that the parameters that weren't working appeared to feel less agency (e.g., *"I don't know, these sliders didn't really seem to make much difference. Halfway through I sort of tried to dial back the verbosity and creativity and it didn't really seem to do much of anything" (P17)).*

Although the tool-like system did not alter the generated text when the slider and dropdown menu values changed, the ambiguity of the parameter terms (e.g., "Creativity" and "Style") led some participants to perceive themselves as having more control over their process. These participants reacted confidently and positively to the experimental system. For example, P6 stated, *"the more control the user has, then the more confident the user feels about the text."* Others perceived that the generated output did not match their desired style and adopted an experimental or exploratory approach where they became interested in seeing *"what the AI was going to do"* (P17). P1 drew a comparison between using the AI and approaching a synthesizer for the first time, aiming to explore what control he had over the AI rather than trying to write a complete piece: *"Encountering it as a tool, my first thing with any tool—like if this was a music program—is just to fuck around with it."* Both P1 and P4 described ceding control to the AI during their interaction to understand its limits, aiming instead to experiment with the system.

5.2.2 Ownership Found in Conceptual Contributions with Agent. Participants that used the agent-like system (Figure 1b) tended to focus on the importance of ideation and conceptual contributions like plot points, character sketches, and settings as opposed to individual sentences. Some participants, like P13, argued that the agent-like system expanded on what they had previously written, stating, *"I think [the generated text] is based on what I've already written down, and it doesn't spark many new ideas for me to write about."* P11 also described how a stronger collaborator in a project is the one who contributes more "ideas" to the piece, *"I do give*

Wordsworth some idea of what I want, but some ideas definitely come from the AI. That's why it's a cooperation, for sure."

One explanation is that participants who used the agent-like system tended to attribute ownership to conceptual contributions because the presence of an agent-like AI afforded the belief participants had the ability to communicate desired intentions to the AI. After using the agent-like system, P9 emphasized this crucial change in dynamic, stating that *"[When I work with a human collaborator] I'll call him out for stuff, and he'll be like, 'actually, I think that works here because of this reason' and I'll be like, 'oh, you know, that's a great point', [but an AI] can't defend those choices and because of that I would want it to mirror what I do.'"* P9's desire for an AI to defend its choices and discuss the piece with them underscores how participants who used the agent-like system valued conceptual communication when interacting with the agentic AI. However, because the agent-like prototype was only capable of fulfilling requests to generate more text, it is less likely that it was the functionality of the prototype that afforded participants to consider conceptual contributions, but rather the interface metaphor itself. It is notable that this focus on conceptual contributions was not restricted to the desire to communicate concepts back and forth in the AI chat window, but that the agentic metaphor afforded *conceptual ownership* among participants. This suggests that agentic metaphors enabled participants to think conceptually about AI co-writing systems, regardless of the actual capabilities of the system.

6 RQ1 Reflection: Taxonomy of Creative Writers' Agency and Ownership

After identifying several conceptions of agency and ownership in Section 5.1, we created a unified, descriptive taxonomy classification that interfaces with the existing literature on these constructs. We therefore crated the provided taxonomy (Table 3).

We observed three types of agency that participants leveraged when working with AI systems. *Explicit agency* [40], which was instantiated differently in the prototype systems, offered the most potential for visible, modifiable, and fine-grained control mechanisms. This type of agency echoes foundational HCI work on user control over interactive systems [67]. Many of our participants described a sense of *implicit agency* [40], exercised through the influence of their text on the output of the AI model. The conception of control as implicit echoes previous HCI findings on collaborative writing that noted how writers implicitly provided context and direction to each other while collaborating [19]. When implicitly exerting control over AI generated responses with their own writing before soliciting generated text from an AI, writers may draw upon previous collaborative writing experiences. *Writing process agency*, which has been less explored in previous HCI research, emerged as a significant source of control based on a writer's cognizance of their own writing processes. Participants frequently described how having final editorial control over the generated work was a significant point of control, echoing previous findings from Reza et al. that found that writers demanded to have a "final say" over AI co-writing processes [59]. However, our work extends Reza et al.'s findings by suggesting that this desire for a "final say" is part of a larger conception of writing process control, which includes control derived from all alignments between AI co-writing and writers' usual creative writing processes, such as editing as one writes or composing without subconscious writing goals being interrupted.

We also found that participants described three types of ownership: *stylistic*, *conceptual*, and *effort-based*. Our findings extend those of Draxler et al. [20] and Reza et al. [59]. While Draxler et al.'s work described the relationship between authorship and ownership and Reza et al.'s framework introduced a dichotomy of writers' values about content versus form, our results revealed a more nuanced phenomenon specifically related to creative writing. Rather than observing ownership conceptions tied to particular populations, we found conceptions of ownership afforded by different interface metaphors and the interaction dynamics these metaphors solicited. This pattern suggests that conceptual and stylistic ownership may represent universal concerns among creative writers, with variations in ownership perceptions emerging from the affordances created by particular interfaces and their corresponding interface metaphors.

These findings also suggest that Pierce et al.'s theory of psychological ownership [55] holds for AI co-writing. Pierce's framework identified three ways individuals develop conceptions of ownership: (1) energy and labor investment, (2) intimate lived experience with the target, and (3) perception of control over the target. Our results demonstrate a strong correspondence between these theoretical constructs and the empirical patterns observed in AI co-writing. Energy and labor investment directly aligns with our findings on effort-based ownership and corroborates established research on the IKEA effect [48] (i.e., labor investment fosters a sense of ownership). Intimate lived experience with the target was fostered through participants' knowledge about the creative process and how the final product reflected their particular stylistic and content contributions. Perception of control over the target explains the correlation between agency and ownership identified by Draxler et al. [20] that manifested in our study when participants exercised explicit, implicit, or writing process agency during AI co-writing.

7 RQ2 Reflection: Design Recommendations for AI Co-Writing Systems

Below, we offer design recommendations grounded in our observations of participants' conceptions of agency and ownership and the interface metaphors they used. These recommendations are intended as situated interpretations of how designers might extrapolate our findings to intentionally design with particular user conceptions in mind. As the results presented for (RQ2) are exploratory and not immediately generalizable, the relationship between interface metaphors and conceptions of agency or ownership presented is not necessarily causal. Participants' existing AI literacy, prior experience with AI systems, and media exposure may also have shaped their conceptions. For example, popular media often depicts AI as either anthropomorphized or tool-like [22], and prior experience with AI tools led some participants (notably P1) to suggest they were "prompting" the AI even when not using the agentic metaphor. The aim of our study, however, was not to establish a causal link between interface metaphors and particular conceptions but to understand how interface metaphors *afforded* particular conceptions. These affordances manifested across participants with

Table 3: Taxonomy of Agency and Ownership Types for Writers Working with AI Co-Writing Systems

Category	Type	Definition
Agency	**Explicit Agency**	Perception of agency exercised through deliberate interface actions such as triggering AI generation, adjusting parameters, or providing direct instructions via chat, text boxes, or dropdown menus.
	Implicit Agency	Perception of agency exercised through the contextual influence of previously written text on AI output, without direct instruction or interface manipulation.
	Writing Process Agency	Perception of agency maintained by integrating AI contributions into established writing and revision workflows, treating generated content as material to be shaped through editing.
Ownership	**Stylistic Ownership**	Perception of ownership derived from a writer's distinctive voice, language choices, and expressive elements that characterize the text.
	Conceptual Ownership	Perception of ownership based on a writer's contribution of foundational ideas, themes, narrative directions, and creative vision.
	Effort-Based Ownership	Perception of ownership earned through the investment of cognitive effort, time, and sustained attention in developing and refining the text.

diverse backgrounds. The recommendations below represent an inital attempt to understand the implications of these affordances.

While designers of AI interface metaphors have previously focused on how mental models may be informed by interface metaphors [16], our findings suggest that altering where the user expects to feel control over their interaction may afford particular feelings about their interaction. Because we aimed to isolate the effects of the interface metaphor to lay the groundwork for further quantitative research, participants did not have different types of control over each system. Our results found that participants had expectations of control that were afforded by the interface metaphors themselves, even if those expectations were not adequately met. Herein, we describe how these findings can help designers appropriately navigate user expectations of agency and ownership as they design and develop AI co-writing system interfaces.

Design Recommendation 1: Consider the Influence of Interaction Metaphors on Different Points of Control. Our findings underscored how participants found different points of control depending on their approach to writing and the affordances of the interface metaphor within the AI co-writing system they used. Points of control are of particular importance because they can shape a writer's expectations of what a particular writing system can or should be capable of accomplishing. Our findings demonstrated that when a participant established particular points of control and those points did not meet their expectations, they were more likely to treat their interaction with an AI co-writing system as an experimental exercise rather than trying to use it to create an accomplished piece of writing. If a particular point of control is expected based on a specific metaphorical presentation, designers aiming for long-term adoption should ensure that the underlying system meets users' expectations.

Designers should also consider gathering additional information about users' writing processes by presenting them with a particular metaphor. To execute this, designers could try to solicit different points of control or conceptions of ownership depending on the metaphor presented. If a designer wants users to find their point of control while "prompting" the AI, they could present an agentic AI that emphasizes that it will be reading the previous text and providing a summary of what has been read before generating new text. Such a change would emphasize to the user that the previously written text is important input to the AI.

Design Recommendation 2: Design for Conceptual Relationships with Agentic AI Assistants. Participants that used Autowrite emphasized the importance of their conceptual contributions more than the other participants. This may have been due to the AI agents not only being conceptualized as capable of providing suggestions or editing the text, but also receiving higher-order suggestions and conversing about the text conceptually with human-like nuance [72]. Designers of AI co-writing systems should thus consider context when deciding whether or not to place an emphasis on higher-order thinking. Designers seeking a collaborative dynamic, in which the user and AI each add language into the final text without discussion, may opt to avoid agentic metaphors. Suppressing, modifying, or avoiding these metaphors would enable users to consider the language within each sentence they compose rather than thinking of user-written sentences as conceptual fodder for the AI's responses.

Design Recommendation 3: Play Into or Against Expectations of Control When Using Tool-Like Metaphors. Our study found that tool-like metaphors often afforded an expectation of explicit control for participants. These metaphors encouraged participants to look to other parameters, dropdown menus, and sliders as means of controlling the generated text. Therefore, designers may want to ensure that user expectations of control are met if interface parameters are presented. However, even if users cannot identify the relationship between interface controls and generated text, this does not necessarily negatively affect their interaction. For example, users may choose to experiment with an AI or generate unusual text to use in other creative output. Such text would need to be creatively inspiring for writers but need not be text that the writer would include in their work.

8 Limitations and Future Work

This research has several limitations. First, our prototype systems were informed by a review of commercial systems available today. We chose to focus on existing commercial systems rather than research prototypes to ensure our metaphors were representative of systems with which writers may already be familiar. As AI co-writing systems are rapidly evolving, new systems may be released within the next few years that introduce new metaphors.

While each prototype system exemplified a different interface metaphor archetype, they only supported participants in using the AI for one task, i.e., extending a piece of writing using GPT-3.5. We thus had to restrict the functionality available in each system. For instance, Wordsworth's conversational interface was limited to scripted responses rather than true dialogue capabilities, and Autowrite's parameter controls were non-functional. These restrictions enabled participants to focus on the presentation of the interface metaphor rather than the different functions and capabilities of the prototypes. While this decision enabled for a comparison between the affordances of each metaphor, it may have attenuated each metaphor's strength by reducing the interaction dynamics generally associated with each. Future research should thus explore how more ecologically valid, but functionally diverse, metaphorical presentations might play a role in conceptions of perceived agency and ownership. Further, a controlled study is needed to establish a causal relationship between these metaphors and conceptions.

The magic-like interface used in our study did not present any affordances of conceptions of agency and ownership. Future work is thus needed to understand if this is an intrinsic property of magic-like interfaces or if this a product of our study methodology and interface presentations. It would also be fruitful to explore the use an interaction paradigm less common in computational interfaces and more 'magical', like waving a hand to generate text, to see if such paradigms heighten the sense of magic perceived.

Also, recent work suggests that longitudinal studies of a week or more may provide more reliable results for AI creative co-writing studies [8]. Further work should thus explore if our results hold over longer periods of time. As conceptions of agency and ownership may also be influenced by cultural contexts external to the North American, English-language creative writing context we explored, it would be fruitful to determine the degree to which our results hold in cultures with different conceptions of ownership such as East Asia, or when other languages or genres of writing are considered.

9 Conclusion

Our research makes several contributions to our understanding of agency and ownership with AI co-writing systems. A review of commercial AI co-writing systems available today led to the identification of three interface metaphor archetypes (i.e., tool-like, agent-like, magic-like). Through an exploratory study that utilized prototypes of these archetypes, we uncovered nuanced interactions between different interface metaphors and perceptions of agency and ownership. Three types of agency and three types of ownership were identified, along with an understanding of the impact of tool-like metaphor interface controls and agent-like chat interfaces on participants' perceptions and behaviors.

Our research also provides recommendations for designers seeking to leverage interface metaphors to shape users' expectations and interactions, including considerations for different points of control, conceptual relationships with agentic AI assistants, and strategically playing into or against expectations of control with tool-like metaphors. By deepening our understanding of writers' perceptions of agency and ownership, as well as how interface metaphors mediate human-AI collaboration while writing, this work lays the groundwork for designing more effective, satisfying, and considered AI co-writing systems that meaningfully balance user agency and ownership.

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Appendix A: Summary Statistics of the Commercial AI Co-Writing Systems Evaluated

Table A1 provides a summary of the commercial AI co-writing systems that were evaluated during our review, along with the number of systems that supported various writing processes and included different AI interaction interface methods.

Appendix B: Interview Protocol for Exploratory Study

Section 1: Introduction

- Explain the study goals and introduce the interviewer
- Request permission to record

Section 2: Background

(1) Discuss the participant's process and how a system's interface presentation affects them
(2) Inquire about the impact of a system's look and feel on their process and emotions
(3) Discuss your relationship with AI and AI writing tools.

Section 3: General Impressions of Prototype System (35 minutes)

Participants used one prototype system for 15 minutes and then the following questions were asked.

(1) What are your initial impressions of the system?
(2) How, if at all, did your writing process change when using the AI system?
(3) Could you envision yourself or other writers incorporating this system into regular writing practice? Why or why not?
(4) Did you encounter any difficulties while using the system?
(5) How did you decide when to request a generation?

Section 4: Follow-Up Questions on Agency and Ownership (20 minutes)

(1) How did the system support or interfere with your control over the writing process? (Agency)
(2) To what extent did you feel the AI system influenced your writing direction? (Agency)
(3) Does the final product represent your unique style and voice? Why or why not? (Ownership)
(4) Do you feel the final product was co-authored with the AI? Why or why not? (Ownership)

Section 5: Follow-Up Questions on Specific Correlations (20 minutes)

(1) Do you feel you collaborated with the AI on this piece?
  (a) How would controlling the output length affect your perception of collaboration?
  (b) How would less similar text (e.g., more ornate language) impact your view of collaboration?
  (c) In what ways did the generated text surprise you? How did this affect your sense of control?
(2) How would a less familiar writing style or voice from the AI affect your sense of control during the interaction?

Table A1: Summary statistics of the review of interface metaphors in commercial AI co-writing systems.

Target Audience	Systems	Writing Processes Supported			AI Interaction Interface				
		Planning	Translation	Reviewing	Chat	Rich Text Editor	Textbox: Ideation	Textbox: Translation	Persistent Outline
Academic Writing (n = 5)	Blainy, CollegeEssay AI Essay Writer, PerfectEssayWriter, TexteroAI, Wordvice AI	4	4	5	1	3	4	5	0
Creative Writing (n = 10)	AutoCrit, DreamGen, Fictionary, NovelAI, NovelCraft, Novelcrafter, Novlr, ProWritingAid, Squibler, Sudowrite	7	6	8	2	6	4	4	6
Enterprise AI Productivity Suite (n = 2)	Notion, Writer	2	2	2	2	2	2	2	2
General Writing (n = 4)	ChatGPT, Grammarly, Gemini in Docs, Quillbot	4	4	4	1	2	4	4	1
Literacy Support (n = 2)	CoWriter, Read&Write	2	2	2	0	2	0	0	0
Social Media, Marketing, and SEO (n = 22)	Anyword, Article Forge, Copy.AI, Frase, Helloscribe, HyperWrite, Jasper AI, Junia, Koala, OwlyWriter, Peppertype, Postwise, Protagoras.app, Rytr, Semrush, Simplified, Surfer, Texta, WordAssistant, WordHero, Wordtune, Writesonic	21	22	16	9	16	22	22	9